TRUONG VAN TINH

AI Application Engineer

Python · FastAPI · RAG · LLM Integration · Evaluation

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PROFESSIONAL SUMMARY

Software engineer building LLM and RAG applications in Python and FastAPI, backed by 3+ years of software development including production .NET, SQL Server and Keycloak work at Con Cung. Focused on reliable AI features: cited retrieval, validated outputs, abstention and fallbacks, automated evaluation and human review.

3+ yrs	8 / 8	10	1,000
software development	eval queries resolved	FastAPI tests, CI green	review decisions validated

PROJECTS

Retail RAG Assistant PERSONAL PROJECT · Python, FastAPI, TF-IDF, SentenceTransformers, Docker — github.com/tvt987/retail-rag-assistant

- Built cited document retrieval with TF-IDF and cosine similarity over paragraph chunks, with optional SentenceTransformers embeddings.
- Developed FastAPI question, document and health endpoints with low-score abstention and extractive fallback when LLM calls fail; 8 evaluation queries across 2 documents gave 6 correct top-3 matches and 2 correct abstentions, CI green.

Support Ticket AI Triage PERSONAL PROJECT · Python, FastAPI, LLM Integration, SQLite, Docker — github.com/tvt987/support-ticket-ai-triage

- Automated LLM-assisted triage for 5 ticket categories with priority and sentiment validation, urgent-rule overrides and a rules fallback; 5 synthetic fixtures passed, human review protected by a token, masked contact data and no raw ticket text in storage.
- Tested 10 FastAPI HTTP scenarios with mocked LLM responses, 120 schema combinations and 720 urgent-rule cases; verified SQLite persistence for 1,000 review decisions, GitHub Actions CI green on main.

Retail Reliability Lab AI-ASSISTED LAB · FastAPI, Redis, SQLite, Docker Compose, Azure

- Redis cache-aside with a 60-second TTL, invalidation and outage fallback; 22 API and cache tests against real Redis; median SQLite lookup cut from 3.63 ms to 0.005 ms over 200 queries on 100,000 synthetic rows; authored Docker Compose and Azure templates.

Layout-Store CONTRIBUTOR · C#, ASP.NET Core, Dapper, PostgreSQL — github.com/tvt987/Layout-Store

- Contributed 10 ASP.NET Core / PostgreSQL routes for search, CRUD, approval and Excel import, with HTTP handlers wired to stored procedures through async interfaces and dependency injection.

WORK EXPERIENCE

Full Stack Developer at **Con Cung** — Nov 2023 – Present

Vietnam's mother-and-baby retail chain · .NET · AngularJS · SQL Server · Dapper · Keycloak · ERP integration

- Delivered AngularJS and .NET voucher workflows, 7 subscription benefit calculations and a tool for 4 generated code-template types.
- Implemented Keycloak JWT / OAuth 2.0 / OIDC authentication and role-based authorization; supported Docker and IIS deployment.
- Fixed SQL Server / Dapper retrieval defects and verified ERP API request and response mappings.

Frontend Intern at **FPT** — Sep 2023 – Nov 2023

- Developed Angular / JavaScript components, integrated e-commerce REST APIs and corrected mobile UI defects.

SKILLS AND TECHNOLOGIES

AI applications	RAG, Document Retrieval, Data Pipelines, NLP (text retrieval), TF-IDF, Semantic Embeddings, LLM Integration, Prompt Engineering, Output Validation, Evaluation, Abstention and Fallbacks, Human-in-the-loop Workflows
Python & APIs	Python, FastAPI, REST APIs, SQLite, Input Validation
Backend & data	C#, .NET, ASP.NET Core, Dapper, SQL Server, PostgreSQL, Redis, Database Design, Query Optimization
Security	Keycloak, JWT, OAuth 2.0, OpenID Connect, RBAC, Contact-data Masking
Testing & DevOps	Unit Testing, Integration Testing, Docker, CI/CD, GitHub Actions, Git, Logging, Monitoring/Observability, Azure, AWS, IIS
Tools & frontend	ChatGPT, Claude, Cursor, GitHub Copilot, MCP, n8n, Angular, AngularJS, TypeScript, JavaScript, RxJS, React, HTML5, CSS3, Tailwind CSS
Practices	Agile/Scrum, Code Review, Technical Documentation, Problem Solving, Cross-team Collaboration, Communication

EDUCATION — **University of Information Technology (UIT), VNU-HCM** · Information Technology · Graduated 2026

LANGUAGES — **English** Good reading and writing (technical documents); basic spoken communication